

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A — Uninformed Search Algorithms

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GitHub Repository Link: https://github.com/Meh02ajv/ai_course_lab1#

Date Submitted: Saturday, June 20, 2026

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	20%
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

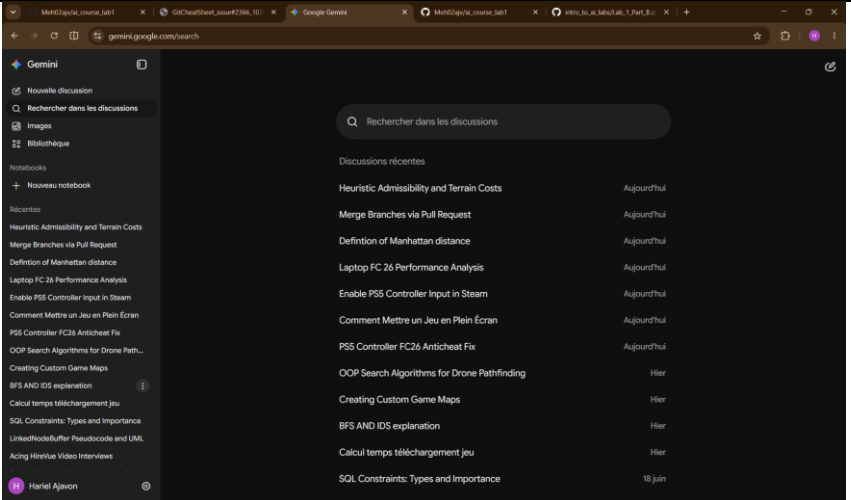
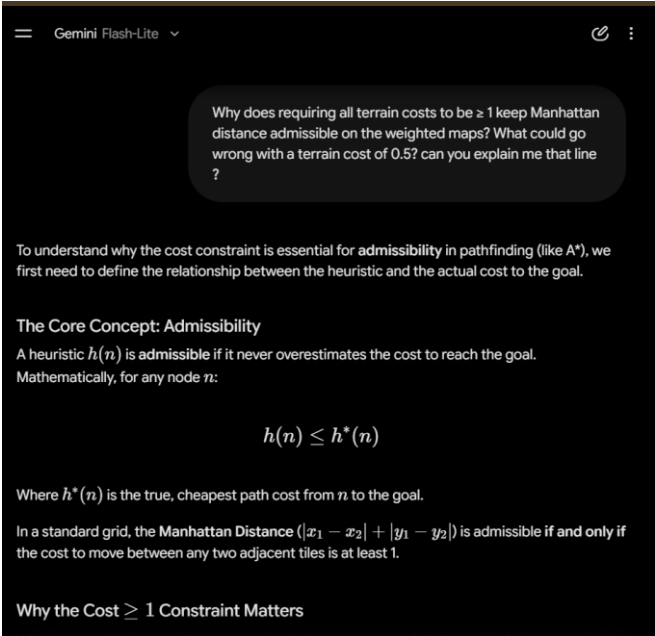
Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Gemini AI	Correction of Errors, Explanation of concepts, Showing the use of syntaxes	- Why does requiring all terrain costs to be ≥ 1 keep Manhattan distance admissible on the weighted maps? What could go wrong with a terrain cost of 0.5? Can you explain that question to me? - Why does reached need to be a dictionary	The question parts and debugging in the TODOs	By referring to the lab instructions and similar implementations on the internet

		(state $\rightarrow$ best node) in this lab, when a plain set was enough for BFS in Part A? • What is Manhattan distance? • Explain me that question: Is Manhattan distance consistent on our unit-cost grid? Check the triangle inequality $h(n) \leq c(n, a, n') + h(n')$ for a single move and explain.		

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	No	
Prompt history	Yes	

		
Screenshot of AI interaction	Yes	

		 Gemini Flash-Lite What is Manhattan distance? Manhattan distance (also known as "Taxicab geometry" or L_1 distance) is a way of measuring the distance between two points on a grid by restricting movement to horizontal and vertical paths only. Instead of measuring the straight-line "as-the-crow-flies" distance (which is Euclidean distance), Manhattan distance simulates moving like a car through city blocks—you must follow the grid lines. The Mathematical Formula For two points on a 2D plane, (x_1, y_1) and (x_2, y_2), the Manhattan distance is the sum of the absolute differences of their coordinates: $d = x_1 - x_2 + y_1 - y_2$ Why it is called "Manhattan" Imagine you are in Manhattan, New York. Because of the rigid grid layout of the streets, you cannot walk diagonally through buildings to get from one corner to another. To go from 1st Demander à Gemini Gemini Flash-Lite Explain me that question: Is Manhattan distance consistent on our unit-cost grid? Check the triangle inequality $h(n) \leq c(n, a, n') + h(n')$ for a single move and explain. To determine if a heuristic is consistent (also called monotonic), we must check if it satisfies the triangle inequality. In pathfinding, consistency means that the heuristic estimate doesn't decrease more than the actual cost of moving between two nodes. Mathematically, it must satisfy: $h(n) \leq c(n, n') + h(n')$ Where: $h(n)$ is the heuristic cost from the current node n to the goal. $c(n, n')$ is the actual cost to move from n to an adjacent node n'. $h(n')$ is the heuristic cost from the new node n' to the goal. Applying it to a Unit-Cost Grid On a grid where every step costs exactly 1 (a unit-cost grid), we are using the Manhattan Demander à Gemini
Exported AI conversation	No	
AI-generated code snippet	Yes	for limit in range(0,max_depth+1): result =dls.search(problem,limit=limit) total_nodes_expanded += result.nodes_expanded

		<pre> max_stack_size = max(max_stack_size, result.max_frontier_size) for child in self.expand(problem, node): if state_is_on_path(node, child.state): continue new_stack_size = current_stack_size + 1 metrics["max_stack_size"] = max(metrics["max_stack_size"], new_stack_size) result, status = self._recursive_dls(problem, child, limit, metrics, new_stack_size) if status == "success": return (result, "success") if status == "cutoff": cutoff_occurred = True return (None, "cutoff" if cutoff_occurred else "failure") </pre>
Revised version showing student input	No	

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: Hariel

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